Gebregziabihier Nigusie Birhane

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Education

Master of Science, Information Technology, Bahir Dar University, Ethiopia **Nov 2020 – Aug 2022**

- Thesis Title: Lexical Complexity Detection and Simplification in Amharic Text Using Machine Learning Approach.
- Honors: Best MSc Thesis Award, First-ranked Thesis in the Department Cohort.

Bachelor of Science , Information Technology, Mizan-Tepi University, Ethiopia. **Oct 2015 – Aug 2019**

- Project: Android Based Tepi Town Crime Report System.

Research Interest

Most large-scale NLP and vision models are trained on high-resource languages and data-rich imaging domains, leaving critical gaps for underrepresented languages and specialized clinical applications. My research targets this gap by fine-tuning pretrained models. Recently, I have also become interested in RAG and multi-agent workflows like LangGraph to build context-aware models, grounding these in responsible AI to ensure the resulting models remain fair and trustworthy.

Research Experience

Graduate Researcher – Master's Thesis, Bahir Dar University, Ethiopia **Sep 2021 – Aug 2022**

- Fine-tuned pretrained language models, such as BERT and RoBERTa, for lexical complexity prediction and simplification in Amharic educational texts.
- Developed foundational complexity detection and simplification models with promising performance.
- Honors: Best MSc Thesis Award, ranked first in the department cohort.

Independent Researcher, Information Technology, Mizan-Tepi University, Ethiopia **Sep 2022 – Present**

- Extended thesis work on Amharic lexical complexity by validating the classification approach (ICAST 2022) and refining the annotation methodology using an automatic annotator tool (IJCAI 2024, AfricaNLP 2025).
- Developed a contour-guided lung segmentation algorithm to validate chest X-ray images, and trained a lightweight mobile deep learning model for pneumonia detection, achieving real-time inference on mobile hardware for low-resource clinical deployment (MICCAI 2025).
- Analyzed sentiment polarity in Amharic-language social media discourse on climate change using LLM, presenting findings at Deep Learning Indaba 2025 and publishing results in PMLR 2026.

Publications

Nigusie, G., Elbatel, M., & Ayana, Y. (2026). Contour-guided segmentation and X-ray image validation for pneumonia detection using mobile deep learning in low-resource settings. In *International Workshop on Medical Image Computing in Resource Constrained Settings (MICCAI 2025)* (pp. 299–309). https://doi.org/10.1007/978-3-032-13654-1_30
- Presented at MICCAI 2025, Daejeon, South Korea.

Nigusie, G., Mossa, N., & Tegegne, T. (2026). Sentiment polarity analysis of Amharic climate change discourse using large language models. *Proceedings of Machine Learning Research*, 1–10. https://proceedings.mlr.press/v302/nigusie26a.html
- Presented at Deep Learning Indaba 2025, Kigali, Rwanda.

Publications (continued)

Nigusie, G. (2025). Supervised machine learning based Amharic text complexity classification using automatic annotator tool. In *Proceedings of the Sixth Workshop on African Natural Language Processing (AfricaNLP 2025)* (pp. 4–14). https://doi.org/10.18653/v1/2025.africanlp-1.2

Nigusie, G., & Tegegne, T. (2024). Applying hybrid approach of complexity annotator tool and machine learning for lexical complexity detection and simplification in Amharic text. In *International Joint Conference on Artificial Intelligence (IJCAI 2024)* (pp. 42–49). https://doi.org/10.24963/ijcai.aai4g.2024/7
- Presented at Deep Learning Indaba 2024, Dakar, Senegal.

Nigusie, G., & Tegegne, T. (2023). Amharic text complexity classification using supervised machine learning. In *Artificial Intelligence and Digitalization for Sustainable Development (ICAST 2022)* (pp. 12–23). https://doi.org/10.1007/978-3-031-28725-1_1
- Presented at CAST 2022, Bahir Dar, Ethiopia.

Nigusie, G., & Tegegne, T. (2022). Lexical complexity detection and simplification in Amharic text using machine learning approach. In *IEEE 4th International Conference on ICT for Development for Africa (ICT4DA 2022)* (pp. 1–6). https://doi.org/10.1109/ICT4DA56482.2022.9971306
- Presented at IEEE ICT4DA 2022, Bahir Dar, Ethiopia.

Projects

Heritage in Code, Wits Innovation Center, South Africa – AI Engineer **Nov 2025 – Jun 2026**

- **Archival Pipeline:** Archived African heritage sounds alongside their cultural context and use.
- **Fusion Engine:** Developed a fusion engine for creative heritage sound reuse, using Chromaprint for audio similarity tracking and Demucs for source separation and pitch analysis.
- **Generative Baseline:** Used MusicGen as a baseline to benchmark heritage in code model's generative outputs. Project detail: https://www.heritageincode.com

Financial Report Generation Agent, Rubani – AI Engineer **Mar 2026 – Present**

- **SQL Generation and Data Retrieval:** Developed a RAG pipeline for vector-based schema ranking and SQL generation from user prompts, retrieving relevant data from relational databases.
- **Report Generation:** Fine-tuned the Qwen-3B model, using RAG-based supervision, to synthesize retrieved tabular records into structured financial reports.

Awards & Grants

- AI and African Music Project Grant ($5,500), Wits University, South Africa, November 2025.
- MICCAI Society MIRASOL Travel Grant (€3,500), Daejeon, South Korea, September 2025.
- Best MSc Thesis Award, Bahir Dar University, Ethiopia, August 2022.

Teaching Experience

- **Lecturer:** Information Technology Department, Mizan-Tepi University, Ethiopia **Aug 2022 – Present** Regularly teach undergraduate courses in Artificial Intelligence and Programming, and advise students on final-year projects. **Assistance Lecturer:** Information Technology Department, Mizan-Tepi University, Ethiopia **Sep 2019 – Nov 2020** Regularly taught undergraduate Lab courses in Programming and advised students on their class projects.

Volunteer

Mentor, Deep Learning Indaba 2024 Mentorship Program **Sep 2024 – Jan 2025**

Volunteer (continued)

- Mentored a team of four students on their hackathon project, providing technical guidance and direction on AI model development, and troubleshooting implementation challenges.

Skills

Programming Languages

- I primarily use Python for research and AI/ML project development.
- I use JavaScript for frontend development, e.g., the Heritage in Code user interface.
- I use LaTeX for everyday document editing, including scientific papers.

ML Tools & Frameworks

- I use PyTorch and TensorFlow for model development and training.
- I use Hugging Face for fine-tuning and hosting pretrained models, e.g., the Heritage in Code fusion engine model.
- I use FastAPI for serving deployed models and connecting them to user-facing interfaces.

Databases

- I commonly use SQL for project data storage.
- I also work with MONGODB and MYSQL.

References

Available on Request